

Withdrawn Teachers and Learned Body Models in Muscle-Driven Motor Learning

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Abstract—A motor learner can be helped by a more skilled teacher or by a model of its own body learned from self-generated movement. Withdrawing a teacher removes its knowledge and changes the learner’s objective at once, so whether withdrawal leaves the learner worse off, and why it drops, have not been separated; for a body model, whether imagined experience helps because of what the model has learned is rarely tested. This paper uses controls that change one thing at a time, on one soft actor-critic learner and six muscle-driven bodies: an elbow and a finger from MyoSuite and four planar arms of one to four joints. A withdrawn teacher is compared with replacing it by a frozen copy of the student, which keeps the imitation term and removes only the teacher; a frozen body model learned from reward-free babbling is compared with a random model in identical imagination machinery. On the finger, a learner whose competent teacher is withdrawn ends ahead of one never guided, and the share of the teacher’s benefit it keeps rises with the time the teacher was attached, from about a quarter after 2,000 steps to about all of it after 7,000. There, after 4,000 or more steps of attachment, most of the drop that follows withdrawal is the removed imitation term rather than the teacher’s not-yet-absorbed knowledge. On planar arms, a random model, whose predictions barely depend on the action, becomes increasingly harmful as joints are added; the babble-trained model avoids this harm.

Index Terms—Developmental robotics, internal models, guidance, musculoskeletal control, reinforcement learning.

I. INTRODUCTION

MUSCLE-DRIVEN bodies are hard to control: each joint is pulled by several muscles, and many activation patterns produce the same movement, so redundancy grows quickly with the number of joints [1]. Deep reinforcement learning on such bodies is slow and is mostly model-free [2]–[4]. Two kinds of help are available to a learner before task learning begins. One is a more skilled other: a teacher policy whose actions the learner imitates. The other is an internal forward model of the body, predicting the consequences of motor commands [5], [6] and learned from self-generated movement [7]; an artificial learner can use it to imagine experience.

Both kinds of help raise a question that existing studies leave open. For the teacher, the question is what happens when it is taken away. Motor-learning research holds that guidance can leave a learner dependent on it [8], [9], and teacher–student pipelines in robotics cut a distillation term to make the student independent [10], [11]. Cutting the term removes the teacher’s knowledge and changes the student’s objective at the same moment, so a collapse at weaning [11] cannot be attributed to either. For the body model, the question is *why* it helps. Imagined rollouts add data whatever the model contains, and

an inaccurate model can make model-based learning worse than model-free [12], [13]; how much of a babble-trained model’s effect is due to what it has learned, and how that depends on the body, has not been isolated.

Both questions can be answered with a control that changes one thing at a time. A frozen copy of the student, swapped in for the teacher at withdrawal, keeps the imitation term and removes only the teacher; comparing it with a plain withdrawal separates lost knowledge from the change of objective. A random model in the identical imagination machinery isolates what the body model has learned.

The study applies these controls to one soft actor-critic learner on six muscle-driven bodies (Fig. 1). The contributions are:

- evidence that a competent teacher can be withdrawn without leaving the learner behind one never guided: on the finger the share of the teacher’s benefit the withdrawn learner keeps rises with the time it was taught, and is more than half on average over the withdrawal times tested (Sec. IV-A);
- a self-anchor control showing that on the finger, after 4,000 or more steps of attachment, most of the drop that follows withdrawal is the removed imitation term rather than the teacher’s not-yet-absorbed knowledge (Sec. IV-B);
- evidence that a body model is worth its content: on a series of planar arms, imagined experience from a random model, whose predictions barely depend on the action, becomes increasingly harmful as joints are added, while the babble-trained model avoids this harm (Sec. IV-C).

The ten primary tests hold under two endpoints and survive a Holm correction (Sec. III-D). Code and data will be released on acceptance.

II. RELATED WORK

Guidance and its withdrawal. The guidance hypothesis [8], [9], [14] holds that frequent augmented feedback guides performance during acquisition but can degrade learning; high-frequency physical guidance gives the poorest retention in one test [15], and reduced knowledge of results improves delayed retention in another [16]. The evidence is mixed: one meta-analysis finds the degrading effect limited to laboratory tasks, parameter learning and extensive practice [17], a later one finds no reliable benefit of reduced feedback frequency [18], and children given reduced feedback learn worse [19]. Performance while guided is not the same as what has been learned [9], [20]. Human studies corrupt or remove the information in

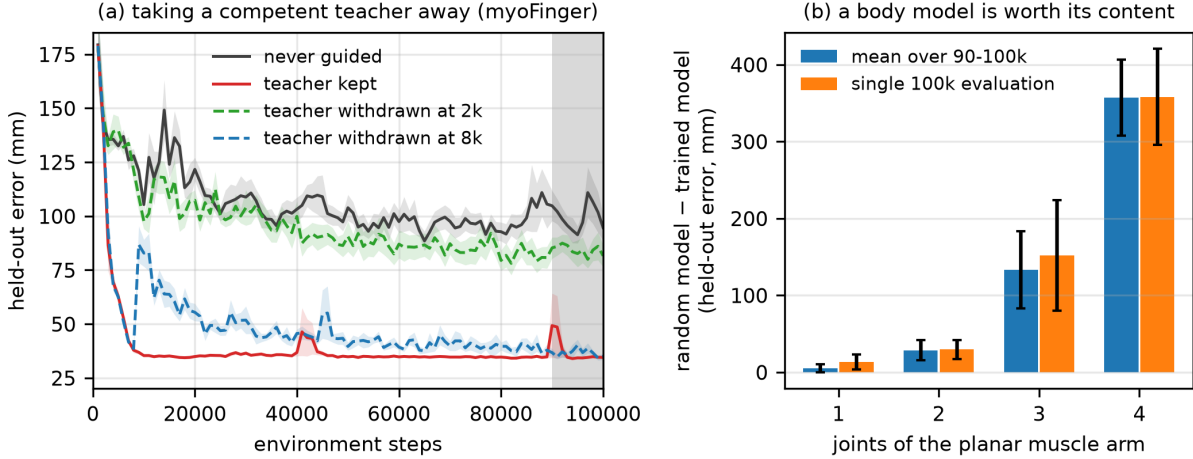


Fig. 1. The two findings. (a) A competent teacher can be taken away: on myoFinger, learners whose teacher is withdrawn (dashed) drop sharply, recover, and end with lower error than a learner never guided (grey), the more so the longer they were guided; mean across 12 seeds, band ± 1 SE, the grey band is the scored window; first evaluation at 1,000 steps. (b) A body model is worth its content: the random model’s error minus the trained model’s error, on planar muscle arms of one to four joints, under both endpoints (bars: mean; whiskers: 95% interval; $n = 12$ on one and two joints, 24 on three and four). The random model’s predictions barely depend on the action; its imagined experience becomes increasingly harmful as joints are added, and the trained model avoids that harm.

a feedback channel while keeping the channel [21]–[23]. In teacher–student learning, a teacher that sees privileged state is distilled into a student on the student’s own rollouts [24]–[27], and the distillation term is annealed or cut to make the student independent [10], [11]. Supervised distillation uses the student itself or a random network as the teacher to separate what the term does from what the teacher knows [28], [29]; a term toward the learner’s own earlier policy stabilises fine-tuning [30]–[32], and a cloned policy fine-tuned by off-policy reinforcement often degrades when the constraint is removed [33]–[35]. What is missing is a comparison of the withdrawn student with one never guided, with the teacher’s knowledge held apart from the change of objective.

Body models. Developmental robotics [36]–[38] treats motor babbling as a way to learn forward models of the self [7], and intrinsic motivation as what drives it [39]; self-modelling [40] and goal babbling [41] learn body models by directed exploration. In model-based reinforcement learning, World Models [42] trains a model once from random rollouts; the Dreamer series [43]–[45] and TD-MPC2 [46] keep the model learning online; MBPO [12] branches short rollouts from real states. Plan2Explore [47] and APV [48] pretrain a model without task reward; offline methods [49]–[51] fit a model to logged, rewarded data. Muscle world models are co-learned with the skill [52], [53], and frozen models learned from reward-free actuation drive soft and muscle-driven bodies by planning or model-only learning [54]–[56]. Model error can make model-based learning worse than model-free, more so in higher dimension [12], [13]. These studies evaluate a model against model-free learning; a comparison with a model that has learned nothing, in the same machinery and across bodies of increasing complexity, is not reported.

III. METHOD

A. Bodies, task and learner

myoElbowPose1D6M is a one-joint elbow driven by six muscles; error is a joint-angle error in milliradians (mrad). myoFingerReachRandom-v0 is a four-joint index finger driven by five muscles; error is a Euclidean fingertip error in millimetres (mm). Both are used with MyoSuite dynamics [57], a reach reward, a fixed start pose and eight training targets. Four planar arms of one, two, three and four revolute joints, driven by two, four, eight and ten MuJoCo muscles (antagonist pairs, biarticular from three joints on), reach a planar fingertip target; error is in mm. Joint count, muscle count and biarticular muscles therefore change together along the arm series. On the one-joint arm the muscles turn the joint only to ± 0.86 rad of the ± 2 rad target range, so 10 of its 20 held-out targets are out of reach and its errors include a floor of about 57.8 mm; across arms only differences between conditions are compared, never absolute errors. Episodes are 100 steps; evaluation is greedy on twenty held-out targets disjoint from the training targets. One soft actor-critic (SAC) [58], [59] learner is used in every condition: two hidden layers of 128 units, update-to-data ratio 2, batch 128, learning rate 3×10^{-4} , automatic entropy temperature. Conditions differ only in what is added. The reward is given on every step in every condition, and every evaluation is made without any help.

B. The teacher and its withdrawal

The teacher is a frozen actor. The student’s actor loss is

$$\mathcal{L}_\pi = \mathcal{L}_{\text{SAC}} + \frac{c}{|\mathcal{A}|} \|\tanh \mu_{\text{student}}(o) - \tanh \mu_{\text{teacher}}(o)\|^2, \quad (1)$$

on replayed observations o , where $|\mathcal{A}|$ is the number of muscles; $c = 1$ while the teacher is attached. The imitation

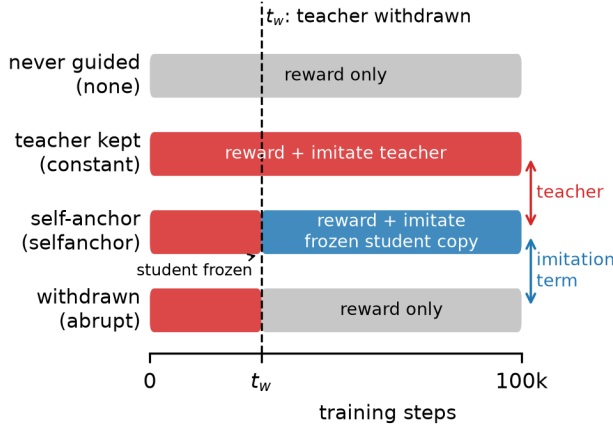


Fig. 2. The teacher-withdrawal conditions. Until t_w every guided learner imitates the teacher (red). At t_w the withdrawn learner loses both the teacher and the imitation term; the self-anchor keeps the term but points it at a frozen copy of the student (blue), so only the teacher is gone. The teacher component is selfanchor – constant, the imitation-term component is abrupt – selfanchor, and the withdrawn learner is compared with one never guided (grey).

term is high-frequency prescriptive feedback about the action, applied to replayed states; the teacher never moves the student in the environment, so it is closer to knowledge of performance than to physical guidance. On the elbow the teacher (T1) is the terminal actor of a model-free run of 30k rewarded steps on a seed outside the study. On the finger, the corresponding actor barely moves (169.1 mm held-out against 172.3 for a zero-activation policy), so the finger teacher (T2) was chosen to be competent: the terminal actor of the best of twelve 100k body-model runs (35.6 mm; the twelve range from 35.6 to 85.3). One student shares its initialisation seed with that run; excluding it changes no conclusion.

At step $t_w \in \{2000, 3000, \dots, 8000\}$ the teacher is withdrawn. *none* is never guided, *constant* keeps the teacher, *abrupt* sets $c = 0$ at t_w , and *selfanchor* replaces the teacher at t_w by a frozen copy of the student’s own actor, with c unchanged for the rest of training (the swap fires two environment steps before t_w , where the weight is last written). The withdrawn learner is compared with *none* to ask whether it is left worse off than a learner never guided. The *drop* at withdrawal is the rise in error over the 1,000 steps after t_w relative to *constant*; the self-anchor splits it (Fig. 2): selfanchor – constant is losing the teacher with the objective kept (the *teacher* component), abrupt – selfanchor is losing the imitation term (the *imitation-term* component). On the elbow the drop is read from a separate set of runs with the same seeds and code, evaluated every 1,000 steps, which reproduce the 100k runs at their shared checkpoints. The snapshot holds what the student has absorbed by t_w , so the teacher component measures only what the teacher still has to add, and it shrinks with attachment partly by construction.

C. The body model and its control

The body model is a forward model $f_\theta(s, a) \mapsto s'$, a two-layer 256-unit network trained once on reward-free Ornstein–Uhlenbeck muscle activations (10^5 transitions on the elbow,

2×10^5 on the other bodies), then frozen. It is used for Dyna [60] rollouts of three steps branched from 200 real states every 100 environment steps; the reward at an imagined state is the environment’s analytic reach reward, computed with the simulator’s forward kinematics. Each update draws 64 real and 64 imagined transitions. Conditions: *prior* uses the trained model; *randprior* uses a randomly initialised network in identical machinery (one random network per body, shared by all seeds, as the trained model is one checkpoint per body), with the trained model’s input and output normalisers (on the one- and four-joint arms those of an earlier build, which differ by at most 0.02 standard deviations). Its predictions barely depend on the action: across the four arms, a change of action moves its predicted joint state 2–4% as much as it moves the trained model’s. *blank* is SAC alone with 128 real transitions per update; *blank64* uses 64 real transitions, each repeated once, the real-data content of a Dyna update without a model.

D. Endpoints and statistics

Each run lasts 100,000 steps and is evaluated every 2,000 steps (every 1,000 in the withdrawal conditions). Final performance is reported under two endpoints: the mean of the six evaluations at 90k–100k (the *late mean*) and the single evaluation at 100k. The late mean was adopted after an internal review because single evaluations of one run move by several mrad or mm between neighbouring checkpoints; claims about final performance are made only when they hold under both. The seed is the unit of analysis: twelve matched seeds per condition, twenty-four on the three- and four-joint arms, paired contrasts, paired- t 95% intervals; bold or * marks an interval excluding zero. A non-detected contrast is not evidence of no difference. The withdrawn learner keeps at least a fraction r of the teacher’s benefit if the one-sided 95% upper bound of (abrupt – none) – r (constant – none), pooled over t_w within each seed, is below zero. The threshold $r = 0.5$, a conventional “more than half”, was chosen during revision, after a descriptive bound had been computed; the largest r that passes is reported as that descriptive bound. Effects of joint count are tested between bodies, with each arm’s seeds independent (Welch t). The primary family, specified during revision, has ten tests: on the finger, the withdrawn-minus-never-guided contrast, retention of at least half the teacher’s benefit and the slope of the withdrawn learner’s advantage over t_w ; and the trained-minus-random gap on the three- and four-joint arms against the one-joint arm; each under both endpoints. All ten remain significant after a Holm correction (largest corrected $p = 0.0018$). The drop at withdrawal is measured from single evaluations; its components are compared per t_w with a Holm correction across the seven t_w of each body.

E. Implementation details

Actor and critics are SiLU networks, the twin critics with layer normalisation; $\gamma = 0.99$, Polyak $\tau = 0.005$, Adam, target entropy $-0.5 |\mathcal{A}|$, gradient norm clipped at 10, real replay 200,000, imagined replay 400,000, 1,000 warm-up steps. Episode time-outs are stored as terminal transitions; imagined transitions are never terminal. Babble data use $\theta = 0.15$,

$\sigma = 0.4$, activations clipped to $[-1, 1]$ and the pose re-randomised every 100 steps. A matched seed sets the network initialisation, replay sampling, the training environment and the order of target draws; runs are deterministic on CPU. The 1,056 runs behind the results take about 1,100 single-thread CPU-hours.

IV. RESULTS

A. A competent teacher can be taken away

On the finger, with a competent teacher, the withdrawn learner ends far ahead of a learner never guided (Table I, Fig. 1a): $-43.25 [-54.72, -31.79]$ mm pooled over the seven withdrawal times (single evaluation: $-40.63 [-53.05, -28.22]$), against a teacher worth $-61.71 [-74.98, -48.44]$. It keeps more the longer it was taught: per withdrawal time the share kept rises from 24% at $t_w = 2000$ to 101% at $t_w = 8000$ (22% to 99% at the single evaluation), and the slope of its advantage over the never-guided learner is $-8.70 [-10.07, -7.33]$ mm per 1,000 attached steps ($-8.44 [-9.89, -6.99]$). Withdrawn at 2,000 or 3,000 steps it keeps less than half (point estimates 24% and 36%); averaged over the seven withdrawal times it keeps more than half (one-sided test against one half, $p = 0.0008$; single evaluation $p = 0.0018$), and at one-sided 95% confidence at least 62% (60% at the single evaluation). From $t_w = 7000$ on it is not detectably behind a learner that kept its teacher, under either endpoint.

On the elbow the teacher is worth less, $-8.65 [-12.25, -5.05]$ mrad, and its benefit goes with it: the withdrawn learner ends 9.2 to 12.2 mrad behind one that kept its teacher at every t_w (7.6 to 17.0 at the single evaluation), and is not detectably different from one never guided ($+1.76 [-1.36, +4.89]$; single evaluation $+3.42 [-5.58, +12.42]$). With a benefit this small, 100k steps cannot exclude a small deficit, so the elbow conditions were run on to 200k on the same seeds (their first 100k reproduce the runs above at every checkpoint). Over 190–200k the withdrawn learner is $+0.04 [-2.11, +2.19]$ mrad from one never guided (single 200k evaluation: $-0.51 [-5.20, +4.18]$); under both endpoints any deficit is smaller than the teacher's full worth, though not smaller than half of it at the single evaluation (one-sided 95%). The never-guided learner is still improving at 200k, so this comparison is not made at a plateau.

B. The drop at withdrawal after longer attachment

Withdrawal is followed by a sharp rise in error (Fig. 1a), of $+42.98 [+8.71, +77.26]$ mm to $+72.54 [+58.26, +86.82]$ on the finger and $+32.05 [+12.70, +51.41]$ mrad to $+80.56 [+23.00, +138.12]$ on the elbow over the next 1,000 steps relative to keeping the teacher (Table II, Fig. 3). On the finger, from $t_w = 4000$ on, the imitation-term component exceeds the teacher component at every t_w , by 17.87 to 46.48 mm, each significant after the Holm correction across t_w ; the imitation term carries 65 to 94% of the drop there (point estimates). On the elbow the teacher component is no longer detected from $t_w = 5000$ on, and the imitation-term component exceeds it significantly after correction at $t_w = 8000$. At $t_w = 2000$ the point estimates favour the teacher component on both bodies

(elbow: by $+32.42 [+3.09, +61.75]$ mrad, uncorrected), but not significantly after correction. The self-anchor keeps the imitation term for the rest of training, so it shows what the drop consists of, not how to avoid it.

C. A body model is worth its content

Table III compares the trained body model with a random model in identical machinery. Because the random model's predictions barely depend on the action, its imagined transitions suggest that actions do not matter; the comparison measures what the trained model's content is worth relative to such uninformative imagination. On the planar arms the gap is $-4.92 [-10.28, +0.43]$ mm on one joint (detected only at the single 100k evaluation), $-28.42 [-41.59, -15.25]$ on two, $-132.89 [-183.14, -82.64]$ on three and $-357.20 [-406.71, -307.69]$ on four (Fig. 1b). Compared between bodies it is larger on three joints than on one by 127.96 mm (single evaluation 138.89) and on four joints than on one by 352.27 mm (344.68), $p < 0.001$ under both endpoints. The trained model also beats the random one on both MyoSuite bodies, by $-25.78 [-34.71, -16.84]$ mrad on the elbow and $-50.11 [-64.54, -35.68]$ mm on the finger.

Most of the growth comes from the random model (Fig. 4). Against model-free learning it is not harmful on one joint ($-4.24 [-6.93, -1.54]$ mm; single evaluation $+4.34 [-3.60, +12.29]$) and increasingly harmful after: $+22.02 [+9.81, +34.23]$ mm on two joints, $+97.37 [+41.17, +153.56]$ on three and $+262.20 [+198.84, +325.57]$ on four. Against model-free learning with the same real data per update (blank64), the trained model's own advantage is not shown to grow with joint count (Table III, last column). The result is therefore not that a correct model is worth more on harder bodies, but that, along the arm series, imagination whose content is uninformative becomes increasingly costly, and the babble-trained model avoids that cost. The MyoSuite bodies do not follow joint count: the random model harms the one-joint elbow ($+21.88 [+14.60, +29.15]$ mrad against model-free learning) and not detectably the four-joint finger ($+3.92 [-14.49, +22.34]$ mm). The gaps are still widening at 100k on the one-, three- and four-joint arms and may be narrowing on two joints ($+9.96 [-0.42, +20.35]$ over the last 20k steps).

V. DISCUSSION

Taking a teacher away. Every evaluation in this study is made without the teacher, so the drop that follows withdrawal is not a retention-test analogue: it is a decline during continued practice once the training signal changes. On the elbow the teacher's benefit does not outlast it, but no deficit relative to never being guided is detected. On the finger a competent teacher's benefit largely persists, the more so the longer the teacher was attached, and the withdrawn learner ends far ahead of one never guided. The self-anchor shows what the drop consists of: on the finger, after longer attachment, it is mostly the removed imitation term, and the teacher's not-yet-absorbed knowledge accounts for less of it. The setting differs from human studies: attachment duration rather than feedback

TABLE I

TAKING THE TEACHER AWAY, POOLED OVER THE SEVEN WITHDRAWAL TIMES ($t_w = 2000, \dots, 8000$), $n = 12$, UNDER BOTH ENDPOINTS. WORTH: CONSTANT—NONE, WHAT KEEPING THE TEACHER GIVES OVER NEVER BEING GUIDED. WITHDRAWN—NEVER: ABRUPT—NONE. KEPT: THE LARGEST FRACTION OF THE TEACHER’S WORTH THAT THE WITHDRAWN LEARNER KEEPS AT ONE-SIDED 95% CONFIDENCE (DESCRIPTIVE BOUND). BOLD: INTERVAL EXCLUDING ZERO.

	late mean (90–100k)			single 100k evaluation		
	worth	withdrawn—never	kept	worth	withdrawn—never	kept
myoElbow (mrad), T1	−8.6 [−12.3, −5.0]	+1.8 [−1.4, +4.9]	−	−8.1 [−15.8, −0.3]	+3.4 [−5.6, +12.4]	−
myoFinger (mm), T2	−61.7 [−75.0, −48.4]	−43.3 [−54.7, −31.8]	≥ 62%	−60.2 [−73.5, −46.8]	−40.6 [−53.0, −28.2]	≥ 60%

TABLE II

THE RISE IN ERROR OVER THE 1,000 STEPS AFTER WITHDRAWAL AT t_w , RELATIVE TO A LEARNER THAT KEEPS ITS TEACHER (TOTAL), SPLIT BY THE SELF-ANCHOR INTO THE TEACHER COMPONENT AND THE IMITATION-TERM COMPONENT; PAIRED MEANS, $n = 12$, *: INTERVAL EXCLUDING ZERO (UNCORRECTED).

		2k	3k	4k	5k	6k	7k	8k
myoElbow (mrad)	total	+59.5*	+39.7*	+63.5*	+44.1*	+80.6*	+32.1*	+45.2*
	teacher	+46.0*	+22.0*	+25.6*	+1.5	+2.7	+7.8	+5.0
	imitation term	+13.5	+17.6*	+37.9*	+42.6	+77.9*	+24.3*	+40.2*
myoFinger (mm)	total	+43.0*	+45.6*	+57.5*	+60.7*	+72.5*	+62.7*	+49.2*
	teacher	+35.7*	+23.6*	+17.2*	+21.4*	+16.1*	+8.1*	+3.0*
	imitation term	+7.3	+22.0	+40.2*	+39.3*	+56.4*	+54.6*	+46.2*

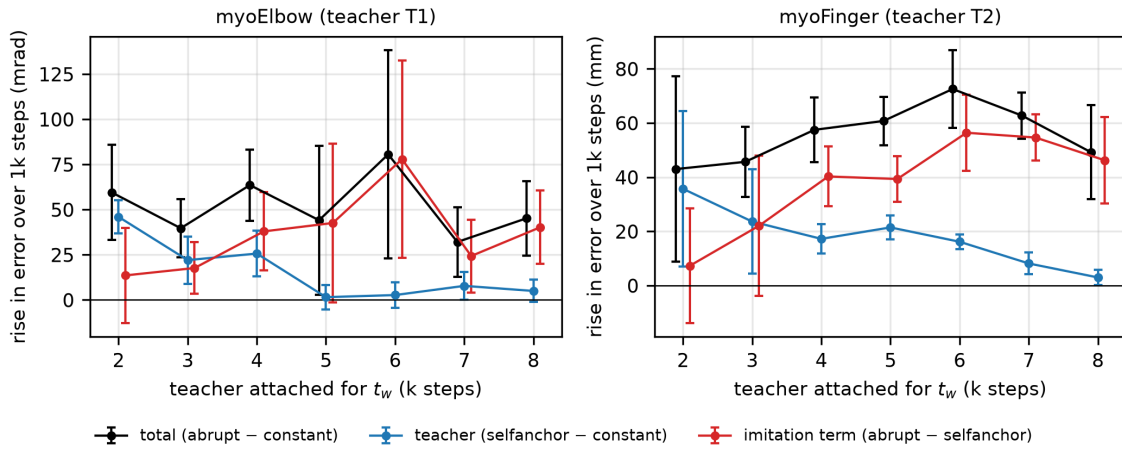


Fig. 3. The drop at withdrawal against the time the teacher was attached (Table II). After longer attachment the drop is mostly the removed imitation term (red); the teacher component (blue) shrinks as the student absorbs the teacher, partly by construction of the self-anchor. Paired means with 95% intervals, $n = 12$.

TABLE III

WHAT THE BODY MODEL HAS LEARNED, ON SIX BODIES: LATE-MEAN ERROR OF THE TRAINED BODY MODEL MINUS A RANDOM MODEL IN IDENTICAL MACHINERY, AND MINUS THE TWO MODEL-FREE BASELINES, WITH 95% INTERVALS. BOLD: INTERVAL EXCLUDING ZERO. †: ALSO EXCLUDING ZERO, WITH THE SAME SIGN, AT THE SINGLE 100K EVALUATION. NEGATIVE FAVOURS THE TRAINED MODEL.

Body	n	trained — random model	trained — blank	trained — blank64
myoElbow (mrad)	12	−25.8 † [−34.7, −16.8]	−3.9 [−7.4, −0.4]	−7.8 [−11.0, −4.6]
myoFinger (mm)	12	−50.1 † [−64.5, −35.7]	−46.2 † [−58.7, −33.7]	−25.9 † [−33.8, −18.0]
1-joint arm (mm)	12	−4.9 [−10.3, +0.4]	−9.2 † [−15.0, −3.3]	−11.7 † [−21.8, −1.7]
2-joint arm (mm)	12	−28.4 † [−41.6, −15.3]	−6.4 † [−11.1, −1.7]	−9.8 † [−16.7, −2.9]
3-joint arm (mm)	24	−132.9 † [−183.1, −82.6]	−35.5 [−62.3, −8.7]	−7.8 [−30.1, +14.4]
4-joint arm (mm)	24	−357.2 † [−406.7, −307.7]	−95.0 † [−134.0, −56.0]	−35.3 † [−59.5, −11.1]

frequency is varied, and reward is given on every step in every condition.

A body model is worth its content. On the planar arms, imagination with a model that ignores the action costs little

on one joint and much on four; with a model learned from babbling, the same machinery helps. What babbling buys is therefore, at least in part, protection against uninformative imagination, and along the arm series that protection matters

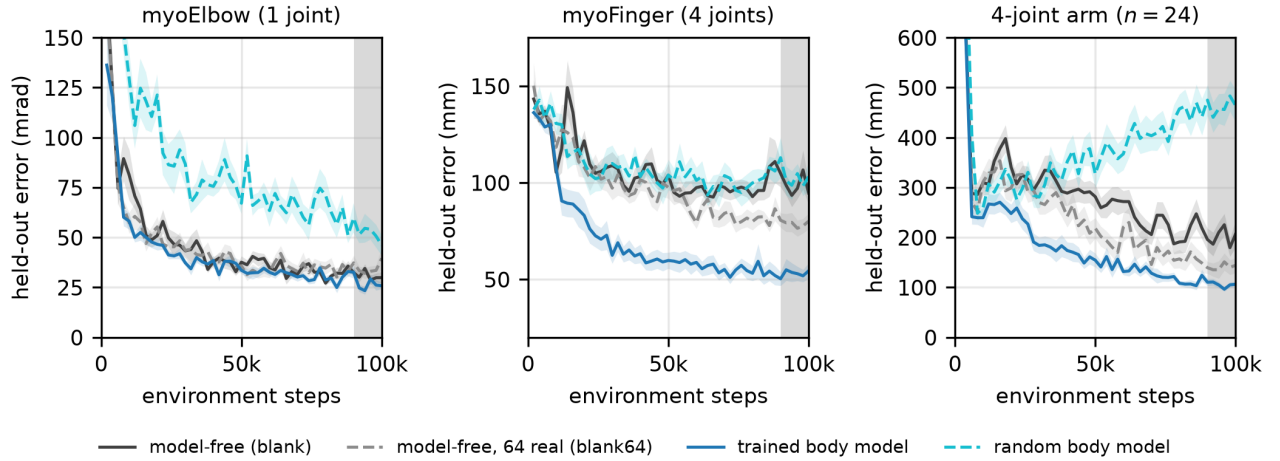


Fig. 4. Learning curves with the trained body model (blue), a random body model in the same machinery (cyan, dashed) and the two model-free baselines (grey). On the four-joint arm the random model ends far above model-free learning while the trained model ends far below it. First evaluation at 2,000 steps. Held-out error, mean, band ± 1 SE; grey band: the scored window.

more as joints are added. Against model-free learning with matched real data, however, the trained model's own advantage is not shown to grow with joint count, so the data do not show that a correct model is worth more on harder bodies.

VI. LIMITATIONS AND FUTURE WORK

The study is single-task reach in simulation, on deterministic, fully observed bodies, which favour a frozen forward model, and the imagined reward uses exact forward kinematics; partially observed bodies and a learned reward are the natural next test. The real replay buffer holds the whole run, so the benefit a withdrawn learner keeps includes guided-phase transitions that remain in its buffer; a buffer purged at withdrawal would separate the two. Each body has one teacher, and the finger's competent teacher is the best of twelve candidate runs; teachers of graded quality would map the benefit against competence. Along the arm series, joint count, muscle count and biarticular muscles change together, and the one-joint arm cannot reach half of its targets; each body has one trained and one random model, shared by all seeds, so variation between model draws is not sampled, and a model that uses the action but has learned the wrong body would be a stronger control. Withdrawal is a single step at 2k–8k steps; gradual fading and withdrawal after more than 8k steps are left to future work, as are longer training on the harder bodies and a physical robot.

VII. CONCLUSION

A teacher and a body model are given to a muscle-driven learner and tested with controls that change one thing at a time. A competent teacher can be taken away: on the finger the share of what the teacher gave that the withdrawn learner keeps rises with the time it was taught, and after longer attachment the drop that follows withdrawal is mostly the removed imitation term. A body model learned from babbling is worth its content: on planar arms, a random model in the same machinery becomes increasingly harmful as joints are added, and the trained model avoids that harm.

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This work made extensive use of Claude (Anthropic), a large language model [61]; every section contains text it generated. Claude wrote and debugged the simulation, training, analysis and figure code; ran and audited the statistical analyses; and drafted and revised the text of every section, including the tables and figure captions. The author directed the study, made the design, analysis and reporting decisions, and reviewed the results, and takes full responsibility for the content of this article.

REFERENCES

- [1] N. A. Bernstein, *The Co-ordination and Regulation of Movements*. Oxford: Pergamon Press, 1967.
- [2] C. H. Berg, V. Caggiano, and V. Kumar, "SAR: Generalization of physiological dexterity via synergistic action representation," in *Proc. Robotics: Science and Systems (RSS)*, 2023.
- [3] A. S. Chiappa, A. Marin Vargas, A. Z. Huang, and A. Mathis, "Latent exploration for reinforcement learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.
- [4] P. Schumacher, D. F. B. Haeufle, D. Büchler, S. Schmitt, and G. Martius, "DEP-RL: Embodied exploration for reinforcement learning in over-actuated and musculoskeletal systems," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2023.
- [5] D. M. Wolpert, Z. Ghahramani, and M. I. Jordan, "An internal model for sensorimotor integration," *Science*, vol. 269, no. 5232, pp. 1880–1882, 1995.
- [6] M. Kawato, "Internal models for motor control and trajectory planning," *Current Opinion in Neurobiology*, vol. 9, no. 6, pp. 718–727, 1999.
- [7] A. Dearden and Y. Demiris, "Learning forward models for robots," in *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2005, pp. 1440–1445.
- [8] A. W. Salmoni, R. A. Schmidt, and C. B. Walter, "Knowledge of results and motor learning: A review and critical reappraisal," *Psychological Bulletin*, vol. 95, no. 3, pp. 355–386, 1984.
- [9] R. A. Schmidt and R. A. Bjork, "New conceptualizations of practice: Common principles in three paradigms suggest new concepts for training," *Psychological Science*, vol. 3, no. 4, pp. 207–217, 1992.
- [10] S. Schmitt, J. J. Hudson, A. Zidek, S. Osindero, C. Doersch, W. M. Czarnecki, J. Z. Leibo, H. Kuttler, A. Zisserman, K. Simonyan, and S. M. A. Eslami, "Kickstarting deep reinforcement learning," arXiv:1803.03835, 2018.

- [11] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Belle-mare, "Reincarnating reinforcement learning: Reusing prior computation to accelerate progress," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [12] M. Janner, J. Fu, M. Zhang, and S. Levine, "When to trust your model: Model-based policy optimization," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [13] C. Voelcker, V. Liao, A. Garg, and A. massoud Farahmand, "Value gradient weighted model-based reinforcement learning," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.
- [14] R. A. Schmidt, D. E. Young, S. Swinnen, and D. C. Shapiro, "Summary knowledge of results for skill acquisition: Support for the guidance hypothesis," *Journal of Experimental Psychology: Learning, Memory, and Cognition*, vol. 15, no. 2, pp. 352–359, 1989.
- [15] C. J. Winstein, P. S. Pohl, and R. Lewthwaite, "Effects of physical guidance and knowledge of results on motor learning: Support for the guidance hypothesis," *Research Quarterly for Exercise and Sport*, vol. 65, no. 4, pp. 316–323, 1994.
- [16] C. J. Winstein and R. A. Schmidt, "Reduced frequency of knowledge of results enhances motor skill learning," *Journal of Experimental Psychology: Learning, Memory, and Cognition*, vol. 16, no. 4, pp. 677–691, 1990.
- [17] F. Marschall, A. Bund, and J. Wiemeyer, "Does frequent augmented feedback really degrade learning? A meta-analysis," *E-Journal Bewegung und Training*, vol. 1, pp. 75–86, 2007.
- [18] B. McKay, J. Hussien, M.-A. Vinh, A. Mir-Orefice, H. Brooks, and D. M. Ste-Marie, "Meta-analysis of the reduced relative feedback frequency effect on motor learning and performance," *Psychology of Sport and Exercise*, vol. 61, p. 102165, 2022.
- [19] K. J. Sullivan, S. S. Kantak, and P. A. Burtner, "Motor learning in children: Feedback effects on skill acquisition," *Physical Therapy*, vol. 88, no. 6, pp. 720–732, 2008.
- [20] N. C. Soderstrom and R. A. Bjork, "Learning versus performance: An integrative review," *Perspectives on Psychological Science*, vol. 10, no. 2, pp. 176–199, 2015.
- [21] M. J. A. Bueckers, R. A. Magill, and K. G. Hall, "The effect of erroneous knowledge of results on skill acquisition when augmented information is redundant," *The Quarterly Journal of Experimental Psychology Section A*, vol. 44, no. 1, pp. 105–117, 1992.
- [22] T. Kitago, S. L. Ryan, P. Mazzoni, J. W. Krakauer, and A. M. Haith, "Unlearning versus savings in visuomotor adaptation: comparing effects of washout, passage of time, and removal of errors on motor memory," *Frontiers in Human Neuroscience*, vol. 7, p. 307, 2013.
- [23] P. Ploettner, C. Muehlberg, F. Psurek, C. Fricke, and J.-J. Rumpf, "Divergent effects of pitch feedback on online and offline motor sequence learning," *Frontiers in Behavioral Neuroscience*, vol. 19, p. 1680277, 2025.
- [24] D. Chen, B. Zhou, V. Koltun, and P. Krähenbühl, "Learning by cheating," in *Proc. Conf. Robot Learning (CoRL)*, 2019.
- [25] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning quadrupedal locomotion over challenging terrain," *Science Robotics*, vol. 5, no. 47, p. eabc5986, 2020.
- [26] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning robust perceptive locomotion for quadrupedal robots in the wild," *Science Robotics*, vol. 7, no. 62, p. eabk2822, 2022.
- [27] A. Kumar, Z. Fu, D. Pathak, and J. Malik, "RMA: Rapid motor adaptation for legged robots," in *Proc. Robotics: Science and Systems (RSS)*, 2021.
- [28] L. Yuan, F. E. H. Tay, G. Li, T. Wang, and J. Feng, "Revisiting knowledge distillation via label smoothing regularization," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [29] F. Sarnthein, G. Bachmann, S. Anagnostidis, and T. Hofmann, "Random teachers are good teachers," in *Proc. Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 202, 2023, pp. 30 022–30 041.
- [30] J. Li, X. Hu, H. Xu, J. Liu, X. Zhan, and Y.-Q. Zhang, "PROTO: Iterative policy regularized offline-to-online reinforcement learning," arXiv:2305.15669, 2023.
- [31] Y. Luo, J. Kay, E. Grefenstette, and M. P. Deisenroth, "Finetuning from offline reinforcement learning: Challenges, trade-offs and practical solutions," arXiv:2303.17396, 2023.
- [32] Y. Zhao, R. Boney, A. Ilin, J. Kannala, and J. Pajarinen, "Adaptive behavior cloning regularization for stable offline-to-online reinforcement learning," in *Proc. European Symp. Artificial Neural Networks (ESANN)*, 2022.
- [33] A. Nair, A. Gupta, M. Dalal, and S. Levine, "AWAC: Accelerating online reinforcement learning with offline datasets," arXiv:2006.09359, 2020.
- [34] I. Uchendu, T. Xiao, Y. Lu, B. Zhu, M. Yan, J. Simon, M. Bennice, C. Fu, C. Ma, J. Jiao, S. Levine, and K. Hausman, "Jump-start reinforcement learning," in *Proc. 40th Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 202, 2023, pp. 34 556–34 583.
- [35] H. Zhang, W. Xu, and H. Yu, "Policy expansion for bridging offline-to-online reinforcement learning," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2023.
- [36] M. Lungarella, G. Metta, R. Pfeifer, and G. Sandini, "Developmental robotics: a survey," *Connection Science*, vol. 15, no. 4, pp. 151–190, 2003.
- [37] M. Asada, K. Hosoda, Y. Kuniyoshi, H. Ishiguro, T. Inui, Y. Yoshikawa, M. Ogino, and C. Yoshida, "Cognitive developmental robotics: a survey," *IEEE Transactions on Autonomous Mental Development*, vol. 1, no. 1, pp. 12–34, 2009.
- [38] A. Cangelosi and M. Schlesinger, *Developmental Robotics: From Babies to Robots*. Cambridge, MA: MIT Press, 2015.
- [39] P.-Y. Oudeyer and F. Kaplan, "What is intrinsic motivation? A typology of computational approaches," *Frontiers in Neurobotics*, vol. 1, p. 6, 2007.
- [40] J. Bongard, V. Zykov, and H. Lipson, "Resilient machines through continuous self-modeling," *Science*, vol. 314, no. 5802, pp. 1118–1121, 2006.
- [41] M. Rolf, J. J. Steil, and M. Gienger, "Goal babbling permits direct learning of inverse kinematics," *IEEE Transactions on Autonomous Mental Development*, vol. 2, no. 3, pp. 216–229, 2010.
- [42] D. Ha and J. Schmidhuber, "Recurrent world models facilitate policy evolution," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018.
- [43] D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi, "Dream to control: Learning behaviors by latent imagination," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.
- [44] D. Hafner, T. Lillicrap, M. Norouzi, and J. Ba, "Mastering Atari with discrete world models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.
- [45] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse control tasks through world models," *Nature*, vol. 640, no. 8059, pp. 647–653, 2025.
- [46] N. Hansen, H. Su, and X. Wang, "TD-MPC2: Scalable, robust world models for continuous control," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024.
- [47] R. Sekar, O. Rybkin, K. Daniilidis, P. Abbeel, D. Hafner, and D. Pathak, "Planning to explore via self-supervised world models," in *Proc. 37th Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 119, 2020, pp. 8583–8592.
- [48] Y. Seo, K. Lee, S. L. James, and P. Abbeel, "Reinforcement learning with action-free pre-training from videos," in *Proc. 39th Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 162, 2022, pp. 19 561–19 579.
- [49] R. Kidambi, A. Rajeswaran, P. Netrapalli, and T. Joachims, "MOREL: Model-based offline reinforcement learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020.
- [50] T. Yu, G. Thomas, L. Yu, S. Ermon, J. Y. Zou, S. Levine, C. Finn, and T. Ma, "MOPO: Model-based offline policy optimization," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020.
- [51] T. Yu, A. Kumar, R. Rafailov, A. Rajeswaran, S. Levine, and C. Finn, "COMBO: Conservative offline model-based policy optimization," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.
- [52] Y. Feng, X. Xu, and L. Liu, "MuscleVAE: Model-based controllers of muscle-actuated characters," in *Proc. SIGGRAPH Asia Conference Papers*, 2023.
- [53] M. Kim and Y. Lee, "FreeMusco: Motion-free learning of latent control for morphology-adaptive locomotion in musculoskeletal characters," in *Proc. SIGGRAPH Asia Conference Papers*, 2025.
- [54] U. Berdica, M. Jackson, N. E. Veronese, J. Foerster, and P. Maiolino, "Reinforcement learning controllers for soft robots using learned environments," in *Proc. IEEE Int. Conf. Soft Robotics (RoboSoft)*, 2024, pp. 933–939.
- [55] T. Wu, S. Li, C. Lyu, K.-W. Sou, W.-S. Chan, and W. Ding, "MoDex: Planning high-dimensional dexterous control via learning neural internal models," arXiv:2409.10983, 2024.
- [56] H. Zheng, B. Sukhija, C. Li, K. Iten, A. Krause, and R. K. Katzschmann, "Learning soft robotic dynamics with active exploration," arXiv:2510.27428, 2025.
- [57] V. Caggiano, H. Wang, G. Durandau, M. Sartori, and V. Kumar, "MyoSuite: A contact-rich simulation suite for musculoskeletal motor control," in *Proc. 4th Annual Learning for Dynamics and Control Conf. (LADC)*, ser. PMLR, vol. 168, 2022, pp. 492–507.

- [58] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in *Proc. 35th Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 80, 2018, pp. 1861–1870.
- [59] T. Haarnoja, A. Zhou, K. Hartikainen, G. Tucker, S. Ha, J. Tan, V. Kumar, H. Zhu, A. Gupta, P. Abbeel, and S. Levine, “Soft actor-critic algorithms and applications,” arXiv:1812.05905, 2018.
- [60] R. S. Sutton, “Dyna, an integrated architecture for learning, planning, and reacting,” *ACM SIGART Bulletin*, vol. 2, no. 4, pp. 160–163, 1991.
- [61] Anthropic, “Claude Opus 5.5,” Large language model, <https://www.anthropic.com/claude>, 2026.